A Practical Guide to Semi-supervised learning

TOPIC -- SEMI-SUPERVISED LEARNING

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argmin f ∈ H (1 l ∑ i = 1 l V (f (x i) , y i) + λ A ■ f ■ H 2 + λ I ∫ M ■ ∇ M f (x) ■ 2 d p (x)) {\displaystyle {\underset {f\in {\mathcal {H}}}{\operatorname {argmin} }}\left({\frac {1}{l}}\displaystyle \sum _{i=1}^{l}V(f(x_{i}),y_{i})+\lambda _{A}\|f\|_{\mathcal {H}}^{2}+\lambda _{I}\int _{\mathcal {M}}\|\nabla _{\mathcal {M}}f(x)\|^{2}dp(x)\right)}

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Heuristic approaches Some methods for semi-supervised learning are not intrinsically geared to learning from both unlabeled and labeled data, but instead make use of unlabeled data within a supervised learning framework. For instance, the labeled and unlabeled examples x 1 , … , x l + u {\displaystyle x_{1},\dots ,x_{l+u}} may inform a choice of representation, distance metric, or kernel for the data in an unsupervised first step. Then supervised learning proceeds from only the labeled examples. In this vein, some methods learn a low-dimensional representation using the supervised data and then apply either low-density separation or graph-based methods to the learned representation. Iteratively refining the representation and then performing semi-supervised learning on said representation may further improve performance. Self-training is a wrapper method for semi-supervised learning. First a supervised learning algorithm is trained based on the labeled data only. This classifier is then applied to the unlabeled data to generate more labeled examples as input for the supervised learning algorithm. Generally only the labels the classifier is most confident in are added at each step. In natural language processing, a common self-training algorithm is the Yarowsky algorithm for problems like word sense disambiguation, accent restoration, and spelling correction. Co-training is an extension of self-training in which multiple classifiers are trained on different (ideally disjoint) sets of features and generate labeled examples for one another.

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Generative models Generative approaches to statistical learning first seek to estimate p (x | y) {\displaystyle p(x|y)} , the distribution of data points belonging to each class. The probability p (y | x) {\displaystyle p(y|x)} that a given point x {\displaystyle x} has label y {\displaystyle y} is then proportional to p (x | y) p (y) {\displaystyle p(x|y)p(y)} by Bayes' rule. Semi-supervised learning with generative models can be viewed either as an extension of supervised learning (classification plus information about p (x) {\displaystyle p(x)}) or as an extension of unsupervised learning (clustering plus some labels). Generative models assume that the distributions take some particular form p (x | y , θ) {\displaystyle p(x|y,\theta)} parameterized by the vector θ {\displaystyle \theta } . If these assumptions are incorrect, the unlabeled data may actually decrease the accuracy of the solution relative to what would have been obtained from labeled data alone. However, if the assumptions are correct, then the unlabeled data necessarily improves performance. The unlabeled data are distributed according to a mixture of individual-class distributions. In order to learn the mixture distribution from the unlabeled data, it must be identifiable, that is, different parameters must yield different summed distributions. Gaussian mixture distributions are identifiable and commonly used for generative models. The parameterized joint distribution can be written as p (x , y | θ) = p (y | θ) p (x | y , θ) {\displaystyle p(x,y|\theta)=p(y|\theta)p(x|y,\theta)} by using the chain rule. Each parameter vector θ {\displaystyle \theta } is associated with a decision function f θ (x) = argmax y p (y | x , θ) {\displaystyle f_{\theta }(x)={\underset {y}{\operatorname {argmax} }}\ p(y|x,\theta)} . The parameter is then chosen based on fit to both the labeled and unlabeled data, weighted by λ {\displaystyle \lambda } :

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History The heuristic approach of self-training (also known as self-learning or self-labeling) is historically the oldest approach to semi-supervised learning, with examples of applications starting in the 1960s. The transductive learning framework was formally introduced by Vladimir Vapnik in the 1970s. Interest in inductive learning using generative models also began in the 1970s. A probably approximately correct learning bound for semi-supervised learning of a Gaussian mixture was demonstrated by Ratsaby and Venkatesh in 1995.

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Continuity / smoothness assumption Points that are close to each other are more likely to share a label. This is also generally assumed in supervised learning and yields a preference for geometrically simple decision boundaries. In the case of semi-supervised learning, the smoothness assumption additionally yields a preference for decision boundaries in low-density regions, so few points are close to each other but in different classes.

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Low-density separation Another major class of methods attempts to place boundaries in regions with few data points (labeled or unlabeled). One of the most commonly used algorithms is the transductive support vector machine, or TSVM (which, despite its name, may be used for inductive learning as well). Whereas support vector machines for supervised learning seek a decision boundary with maximal margin over the labeled data, the goal of TSVM is a labeling of the unlabeled data such that the decision boundary has maximal margin over all of the data. In addition to the standard hinge loss (1 − y f (x)) + {\displaystyle (1-yf(x))_{+}} for labeled data, a loss function (1 − | f (x) |) + {\displaystyle (1-|f(x)|)_{+}} is introduced over the unlabeled data by letting y = sign ■ f (x) {\displaystyle y=\operatorname {sign} {f(x)}} . TSVM then selects f ∗ (x) = h ∗ (x) + b {\displaystyle f^{*}(x)=h^{*}(x)+b} from a reproducing kernel Hilbert space H {\displaystyle {\mathcal {H}}} by minimizing the regularized empirical risk:

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argmax Θ (log ■ p ({ x i , y i } i = 1 l | θ) + λ log ■ p ({ x i } i = l + 1 l + u | θ)) {\displaystyle {\underset {\Theta }{\operatorname {argmax} }}\left(\log p(\{x_{i},y_{i}\}_{i=1}^{l}|\theta)+\lambda \log p(\{x_{i}\}_{i=l+1}^{l+u}|\theta)\right)}

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An exact solution is intractable due to the non-convex term (1 − | f (x) |) + {\displaystyle (1-|f(x)|)_{+}} , so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).